

¹ Graphical Abstract

² **Bias against Einstein data in the construed hypothesis**

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4 Highlights

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- 7 • Research highlight 1 Entangled polarization of photons.
- 8 • Research highlight 2 Misconstruction of the hypotheses behind the Bell
- 9 correlation formula

10 Bias against Einstein data in the construed hypothesis

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12 Abstract

In the paper it is demonstrated that, given Alice and Bob hold equal copies of measuring instruments, the claim that theoretically the quantum correlation can be written as a Bell correlation is refuted. The hypothesis that theoretically the quantum correlation in entangled polarized photons is misconstrued.

13 *Keywords:* Entangled photons, Bell's correlation formula, equal
14 measurement functions, misconstruction of hypotheses

15 1. Introduction

16 It all starts with Einstein not agreeing with the completeness of quan-
17 tum mechanics. That happened already early in the first stages of the the-
18 ory (Howard, 1990). Later Einstein developed a more detailed version of
19 his criticism and invented what nowadays is called entanglement (Einstein
20 et al., 1935). Later Einstein wrote a less "Gelehrter" see a.o. (Howard,
21 1990), (Butzer et al., 2025), version that better suited his philosophical taste,
22 (A.Einstein, 1948). In the later stages of Einstein's thinking about quantum
23 mechanics (Butzer et al., 2025), Einstein in a letter to Lionel Cooper, clearly
24 distinguished between inner contradiction and incompleteness.

25 The problem with the claim that there must be hidden variables (that
26 restore causality and locality) is that no example or direction to search are
27 given by Einstein. Einstein leaves it to next generations to figure out a local
28 hidden variables theory of quantum entanglement.

29 In the fifties, Aharonov and Bohm simplified the notion of entanglement
30 to spin entanglement (Bohm and Aharonov, 1957). Later, John Bell em-
31 ployed the spin entanglement in his formulation of the spin (polarization)
32 entanglement correlation formula (Bell, 1964).

33 Bell's correlation formula is fully embedded in classical probability in
 34 the sense that there is a space of ancillary (Einsteinian) hidden variables λ
 35 in a universe set Λ with a density that has all characteristics of a classical
 36 probability density $\rho(\lambda) \geq 0$; see equation (1) below. At a later stage Aspect
 37 et al construed a paradigmatic experiment out of Bell's concept (Aspect et al.,
 38 1982).

39 If the classical probability or Kolmogorovian nature of the correlation
 40 formula of Bell's correlation formula is denied, then the problem was solved
 41 by o.a. the present author begin 2000 (H.Geurdes, 2001). Furthermore,
 42 in a recent paper, the present author demonstrated that Bell's correlation
 43 formula, looking at the Krein-Milman theorem from measure theory in the
 44 form of a Pettis integral, is an incomplete description of the empirical reality,
 45 see (Geurdes, 2026b). The paper before you is an attempt to simplify and
 46 strengthen the result in (Geurdes, 2026b).

47 2. Math

48 2.1. Hypotheses

49 Bell's correlation $E(\mathbf{a}, \mathbf{b})$, between unit length parameter vector $\mathbf{a}$ of Al-
 50 ice's measurement function denoted with $A(\mathbf{a}, \lambda) = \pm 1$ and unit length pa-
 51 rameter vector $\mathbf{b}$ of Bob's measurement function $A(\mathbf{b}, \lambda) = \pm 1$ is defined
 52 by

$$53 \quad E(\mathbf{a}, \mathbf{b}) = \int_{\lambda \in \Lambda} \rho(\lambda) d\lambda A(\mathbf{a}, \lambda) A(\mathbf{b}, \lambda) \quad (1)$$

54 In (1) the situation is modeled where Alice and Bob have the same type of
 55 measuring instrument, mathematically described by function $A(\mathbf{x}, \lambda)$. Alice
 56 has $\mathbf{x} = \mathbf{a}$ and Bob has $\mathbf{x} = \mathbf{b}$, the λ are hidden in "the system." The set
 57 of (Euclidean) unit length $\mathbb{R}^3$ vectors is denoted with $\mathcal{S}$. The $\lambda \in \Lambda$ are
 58 Einsteinian ancillary variables, and $\Lambda \subseteq \mathbb{R}^n$ is the probabilistic universe set
 59 of the ancillary variables. If we denote $\lambda(dP_\rho) = \rho(\lambda)d\lambda$ as the probability
 60 measure, then the complete classical probability space is $(\Lambda, \Sigma, P_\rho)$. Here,
 61 Σ is a set of subsets of the universe set Λ such that it is a sigma-algebra
 62 (Rosenthal, 2008). The physical nature of the λ is unknown. The $\rho(\lambda) \geq 0$
 63 for all $\lambda \in \Lambda$, is the (probability) density of the ancillary variables. Therefore,
 64 we have, $\int_{\lambda \in \Lambda} \rho(\lambda) d\lambda = 1$ and for $\Lambda' \subset \Lambda$, the probability is therefore, $P_\rho(\Lambda') =$
 65 $\int_{\lambda \in \Lambda'} \rho(\lambda) d\lambda \in [0, 1]$.

66 In order to make the possibility of ancillary Einstein variables λ in the
67 system accessible to experiment, Bell formulated a correlation (expectation
68 value) see (1) to obtain the quantum correlation between entangled pairs
69 of photons (or positron-electron pairs or..). In this spirit, it is possible to
70 formulate the following two hypotheses for circular polarized photon pairs
71 (Shih et al., 1995). Here,

$$H_0 : \mathbf{x} \cdot \mathbf{y} = \cos(\psi_{x,y}) = \int_{\lambda \in \Lambda} \rho(\lambda) A(\mathbf{x}, \lambda) A(\mathbf{y}, \lambda) d\lambda$$

72

H_1 : the hypothesis H_0 is false

73 The H_0 hypothesis implies that there are $\lambda \in \Lambda$ and there is at least one
74 density $\rho(\lambda) \geq 0$, with $\int_{\lambda \in \Lambda} \rho(\lambda) d\lambda = 1$, and at least one form of measurement
75 function $A(\mathbf{x}, \lambda)$ exist in empirical reality, such that the quantum correlation
76 $\mathbf{x} \cdot \mathbf{y} = \cos(\psi_{x,y})$ for all possible symmetrical combinations of $\mathbf{x}$ and $\mathbf{y}$ in $\mathcal{S}$,
77 can be obtained from (1). In other words, it is possible to have in empirical
78 reality that H_0 is true. If that is denied, then postulating the correlation
79 based on ancillary λ etc, in (1) makes no sense at all. Why work with
80 a classical probability correlation equivalent of the quantum correlation if
81 some symmetrical combinations of $\mathbf{x}$ and $\mathbf{y}$ in $\mathcal{S}$ are excluded beforehand to
82 have H_0 is true?

83 Hypothesis H_1 can be true in a number of different ways. It can be true
84 by having gathered data and the finding of the data is such that $\cos(\psi_{a,b})$ is
85 not equal to $E(\mathbf{a}, \mathbf{b})$ in (1). Note that there is an essential difference between,
86 "there are symmetrical combinations of $\mathbf{x}$ and $\mathbf{y}$ in $\mathcal{S}$ that, beforehand do
87 not have H_0 " and "we found in the data that H_0 is not true."

88 Or, H_1 can be true on logical grounds. In a paper of the present au-
89 thor, (Geurdes, 2026a) it was demonstrated that the conclusion based on
90 the gathering of data in Aspect's experiment was statistically flawed. Hence,
91 Aspect's claim that H_1 was proven because of the data found in the experi-
92 ment, is false. The hypothesis construed by Aspect et al in the paradigmatic
93 experiment was a misconstruction that was biased against finding Einstein
94 data in the empirical reality. Why did that occur.

95 2.2. Inequality

96 In this section we will try to find out if the hypothesis H_0 is well-formed.
97 In a way that is employed in deriving all kinds of inequalities, we can employ

the definition in (1) to obtain

$$E(\mathbf{a}, \mathbf{b}) - E(\mathbf{a}, \mathbf{d}) = \int_{\lambda \in \Lambda} d\lambda \rho(\lambda) A(\mathbf{a}, \lambda) A(\mathbf{b}, \lambda) \{1 - A(\mathbf{b}, \lambda) A(\mathbf{d}, \lambda)\} \quad (2)$$

Because, $\rho(\lambda) \geq 0$ and $\{1 - A(\mathbf{b}, \lambda) A(\mathbf{d}, \lambda)\} \geq 0$, while $A(\mathbf{a}, \lambda) A(\mathbf{b}, \lambda) \leq 1$, it is found that

$$E(\mathbf{a}, \mathbf{b}) - E(\mathbf{a}, \mathbf{d}) \leq \int_{\lambda \in \Lambda} d\lambda \rho(\lambda) \{1 - A(\mathbf{b}, \lambda) A(\mathbf{d}, \lambda)\} \quad (3)$$

Because, $A(\mathbf{b}, \lambda) A(\mathbf{d}, \lambda) = \pm 1$, we find that for the set $\Lambda_{\neq}(\mathbf{d}, \mathbf{b})$ of λ where $A(\mathbf{d}, \lambda) A(\mathbf{b}, \lambda) = -1$, and $\mathbf{a}, \mathbf{b}$ and $\mathbf{d}$ in $\mathcal{S}$, that,

$$E(\mathbf{a}, \mathbf{b}) - E(\mathbf{a}, \mathbf{d}) \leq 1 - E(\mathbf{d}, \mathbf{b}) \quad (4)$$

and $1 - E(\mathbf{d}, \mathbf{b})$ equal to $2P_{\rho}(\Lambda_{\neq}(\mathbf{d}, \mathbf{b}))$. The probability derives from the probability space is $(\Lambda, \Sigma, P_{\rho})$.

If the hypothesis H_0 is true then for all symmetrical combinations of $\mathbf{a}, \mathbf{b}$ and $\mathbf{d}$ in $\mathcal{S}$ we must have $\cos(\text{angle between } \mathbf{a}, \mathbf{b})$ etc. Or, $\cos(\psi_{a,b}) - \cos(\psi_{a,d}) \leq 1 - \cos(\psi_{d,b})$. Now generally $\cos(\psi) = 1 - 2\sin^2(\frac{1}{2}\psi)$. And so

$$\sin^2\left(\frac{1}{2}\psi_{a,d}\right) - \sin^2\left(\frac{1}{2}\psi_{a,b}\right) - \sin^2\left(\frac{1}{2}\psi_{d,b}\right) \leq 0 \quad (5)$$

A numerical study shows, with code in Appendix A, that inequality (5) is not true for all symmetrical combinations of $\mathbf{a}, \mathbf{b}$ and $\mathbf{d}$ in $\mathcal{S}$. If the reader runs the provided R code, the outcome of the numerical study becomes clear. It is clear from the numerical study that at least one of the angles $\psi_{a,b}, \psi_{a,d}$ or $\psi_{d,b}$, two of them or perhaps all three of them together, are not in accordance with H_0 . To be specific

$$\begin{aligned} & \cos(\psi_{a,d}) \not\equiv \int_{\lambda \in \Lambda} d\lambda \rho(\lambda) A(a, \lambda) A(d, \lambda) \\ & \text{and/or} \\ & \cos(\psi_{a,b}) \not\equiv \int_{\lambda \in \Lambda} d\lambda \rho(\lambda) A(a, \lambda) A(b, \lambda) \\ & \text{and/or} \\ & \cos(\psi_{d,b}) \not\equiv \int_{\lambda \in \Lambda} d\lambda \rho(\lambda) A(d, \lambda) A(b, \lambda) \end{aligned} \quad (6)$$

This is not a matter of "let us check in nature if H_0 is true for all symmetrical combinations of $\{\mathbf{a}, \mathbf{b}, \mathbf{d}\}$." It, H_0 that is, is simply already not true on theoretical/numerical grounds.

3. Conclusion & discussion

We may therefore conclude that it was demonstrated that H_0 is false. But H_0 is not false on grounds of gathered data. One of the angles $\psi_{a,b}$, $\psi_{a,d}$ or $\psi_{d,b}$, two or perhaps all three, are not in accordance with H_0 . It is a misconstrued hypothesis. This is so because the inequality (5) only refers to the formulation of the hypothesis H_0 . The failure of H_0 does not tell us anything about possible Einstein affirmative data at all. It is also not a famous Bell inequality. It is the demonstration of a misconstrued hypothesis because some $\psi_{a,b}$, $\psi_{a,d}$ and/or $\psi_{d,b}$ already destroy the required generality of H_0 . What is the use of a hypothetical correlation when some combinations of unit parameter vectors for sure don't work?

The formulated H_0 can not, from the start of an empirical research, only be considered valid for *some* symmetrical combinations of three different unit length vectors to be found on $\mathcal{S}$. What do you do with the ones that don't comply to H_0 ? The conclusion is therefore that it is not possible with H_0 to cover *all* $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{d}$ in $\mathcal{S}$. And that refers back to the impossibility to have an all encompassing $E(\mathbf{a}, \mathbf{b})$ in (1). And we note that the inequality (5) is derived in the same manner as other Bell inequalities.

Hence, when one uses H_0 and H_1 in an empirical exploration, the misconstruction of H_0 will show itself for equal type of measuring instrument functions, $A(\mathbf{x}, \lambda)$ for both Alice and Bob. If an experimental researcher wants to contest this "equal type of measuring instrument functions" point with an experiment, it first and foremost must be demonstrated that the measurement function of Alice, i.e. A and the measuring instrument function of Bob, i.e. B respond differently to a given settings given a λ . That appears to be impossible because the functions $A(\mathbf{x}, \lambda) = \pm 1$ and $B(\mathbf{y}, \lambda)$ are unknown. With the same type of measuring instrument for Alice and Bob one may, however, initially assume that each measurement instrument will respond similarly to input. And note that the quantum correlation has $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$.

Declarations

The author declares no financial interests and the work was not funded.

Appendix A

R code employed in the numerical study mentioned in the body of the paper.

```

158 #
159 fdrie<-function(){
160   s<-runif(3)
161   s1<-sign(s-0.5)
162   a<-runif(3)
163   norma<-sqrt(sum(a*a))
164   x<-a/norma
165   y<-s1*x
166   return(y)
167 }
168 ntel<-0
169 blStp<-FALSE
170 while (!blStp){
171   ntel<-ntel+1
172   a<-fdrie()
173   b<-fdrie()
174   d<-fdrie()
175   eab<-sum(a*b)
176   ead<-sum(a*d)
177   edb<-sum(d*b)
178   x<-acos(eab)/2
179   y<-acos(ead)/2
180   z<-acos(edb)/2
181   sab<-sin(x)*sin(x)
182   sad<-sin(y)*sin(y)
183   sdb<-sin(z)*sin(z)
184   s<-sad-sab-sdb
185   if(s>0.2){
186     print('***')
187     blStp<-TRUE
188     print(s)
189     print(a)
190     print(b)
191     print(d)
192   }
193   if(ntel>1000){
194     blStp<-TRUE
195   }

```

```

196 }
197 print(ntel)
198 print("* end *")
199

```

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